ChE 451 (Renewable Energy Technologies)
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Group Members: Zainab Ahmed, Diana Awimrin, Leena Daniel, Ariel Maret, Sherin Skariah

Abstract

The new chemical engineering building (EIB) uses electricity to operate the ventilation system in the labs. If there is a power outage, all students and staff are told to exit the building because the labs' ventilation systems are off. Without ventilation, anyone could inhale hazardous chemicals. This project will entail designing a renewable energy microgrid system for the new chemical engineering building (EIB) to provide electricity during power outages.

By gathering research studies on the efficiency of microgrid systems, this project will show the pros and cons of different sources of renewable energy to produce electricity for the microgrid system. A renewable source is chosen to generate electricity. A major concern to be addressed is how this energy must be stored for maximum efficiency under a variety of circumstances.

As part of our research we will obtain utility information for EIB and calculate the amount of electricity required for one day. This amount of energy will be stored in the battery. As part of our research we will investigate whether it is viable to sell any remaining electricity back to the grid.